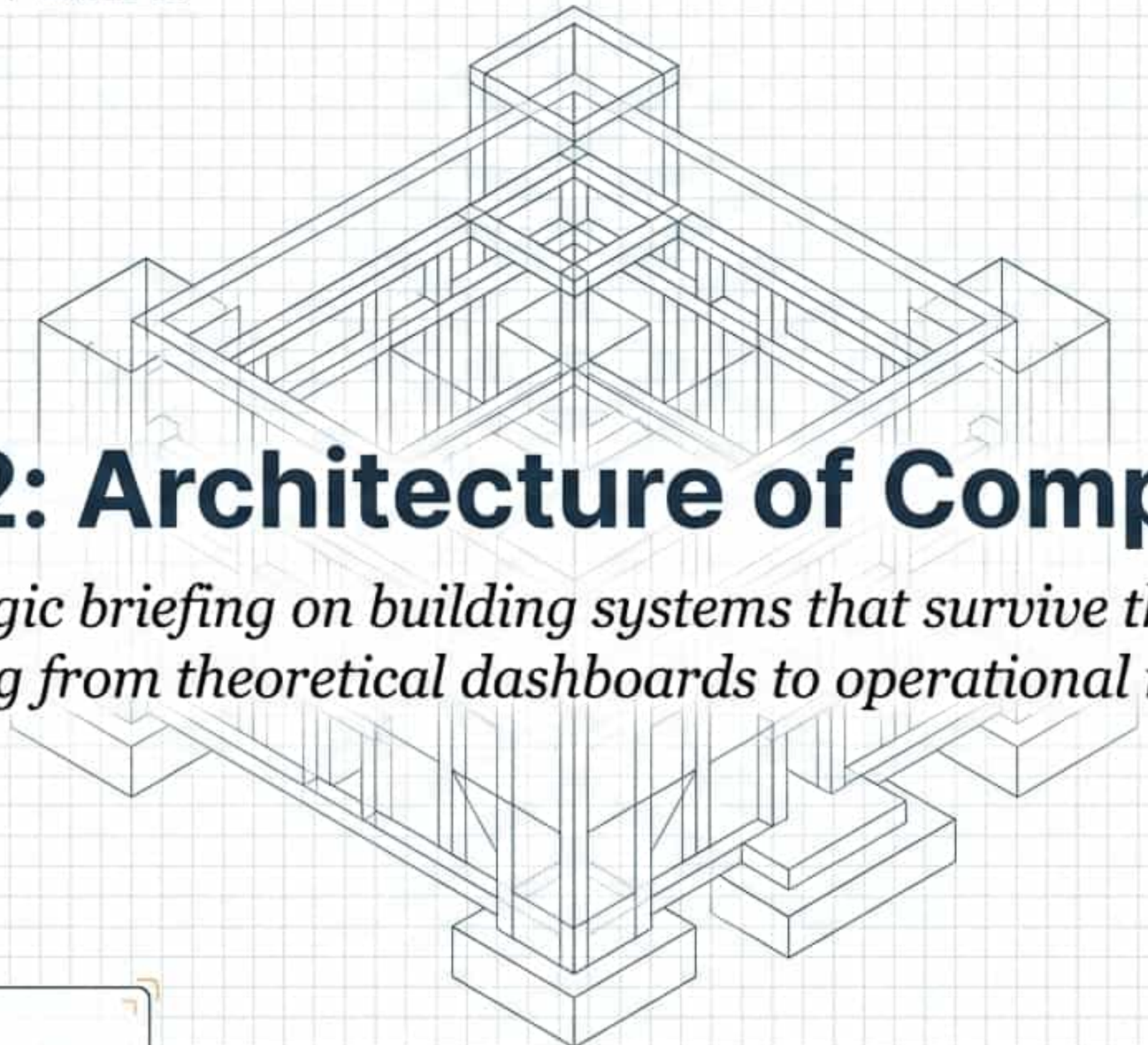

Level 2: Architecture of Compliance

A strategic briefing on building systems that survive the audit.
Moving from theoretical dashboards to operational reality.

EST. TIME: 6-8 HOURS
FOCUS: OPERATIONAL ARCHITECTURE

STATUS: DRAFTING

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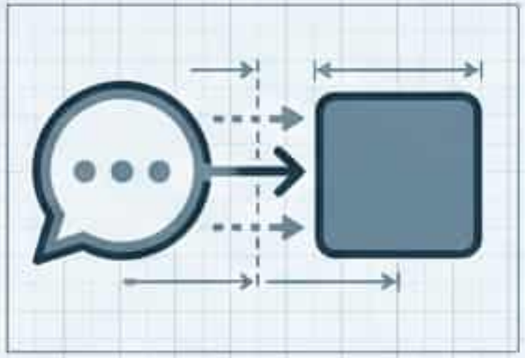
BUILDING SYSTEMS THAT SURVIVE THE AUDIT

Executive Summary: Operational Architecture is the difference between a compliant report and a reputational risk.



1. Translation

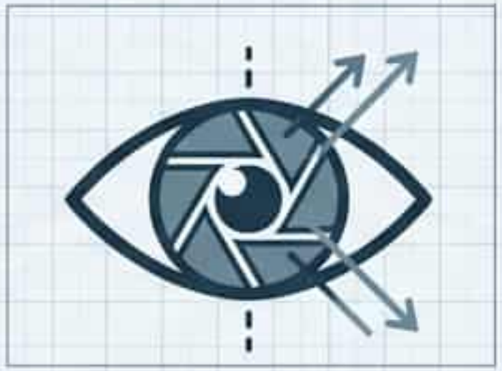
Bridging the gap between framework vocabulary and audit vocabulary.

2. Firewalls

Implementing EU Taxonomy Technical Screening Criteria (TSC).

3. Lineage

Building 'Evidence Ladders' that trace AI insights to source documents.



4. Classification

Balancing transparency with trade secret protection.

Note

CORE THESIS: A SHINY DASHBOARD IS A LIABILITY IF THE ENGINEERING CANNOT WITHSTAND FORENSIC SCRUTINY.

High-Fidelity Architectural Blueprint

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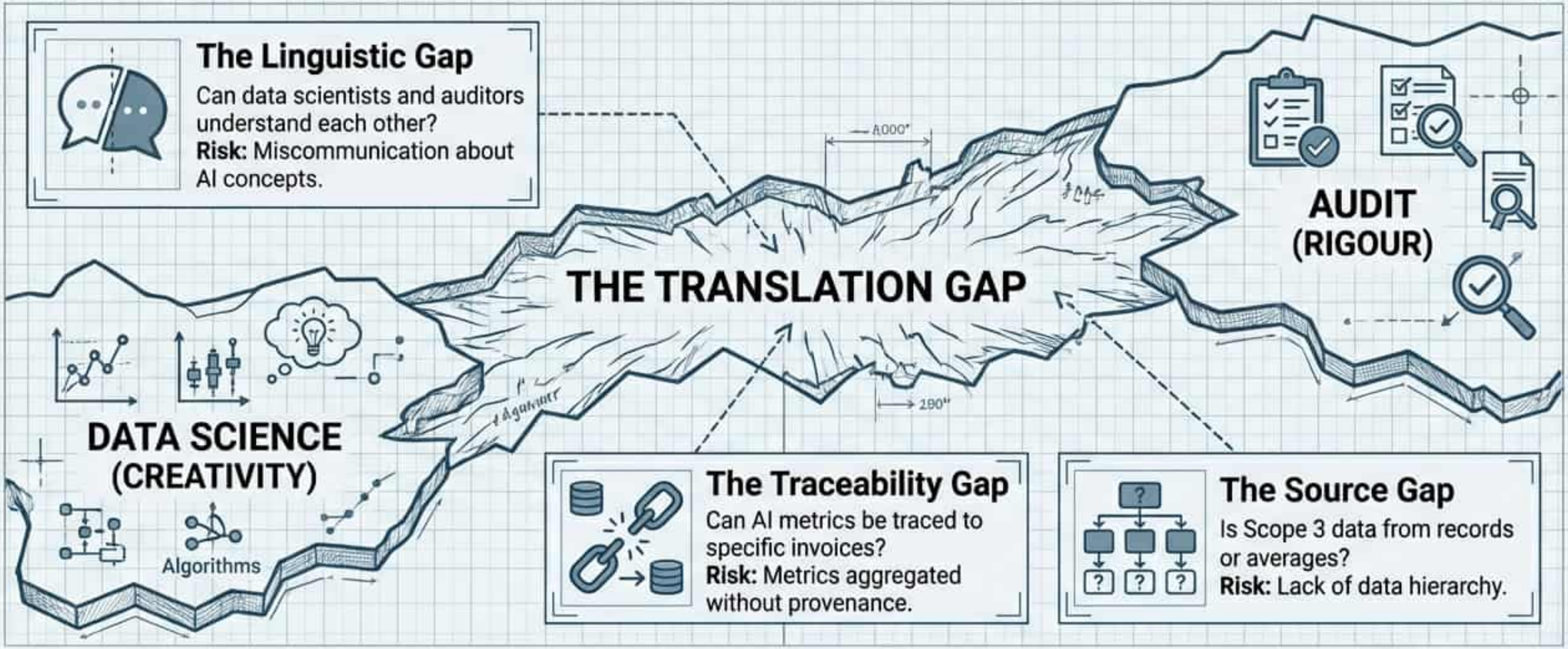
THE OPERATIONAL CHALLENGE: TRANSLATION RISK

Based on Pre-Work Assessment Data



The Linguistic Gap

Can data scientists and auditors understand each other?
Risk: Miscommunication about AI concepts.

The Traceability Gap

Can AI metrics be traced to specific invoices?
Risk: Metrics aggregated without provenance.

The Source Gap

Is Scope 3 data from records or averages?
Risk: Lack of data hierarchy.

Note: Bridging these gaps requires systematic operational architecture, not just better tools.

High-Fidelity Architectural Blueprint

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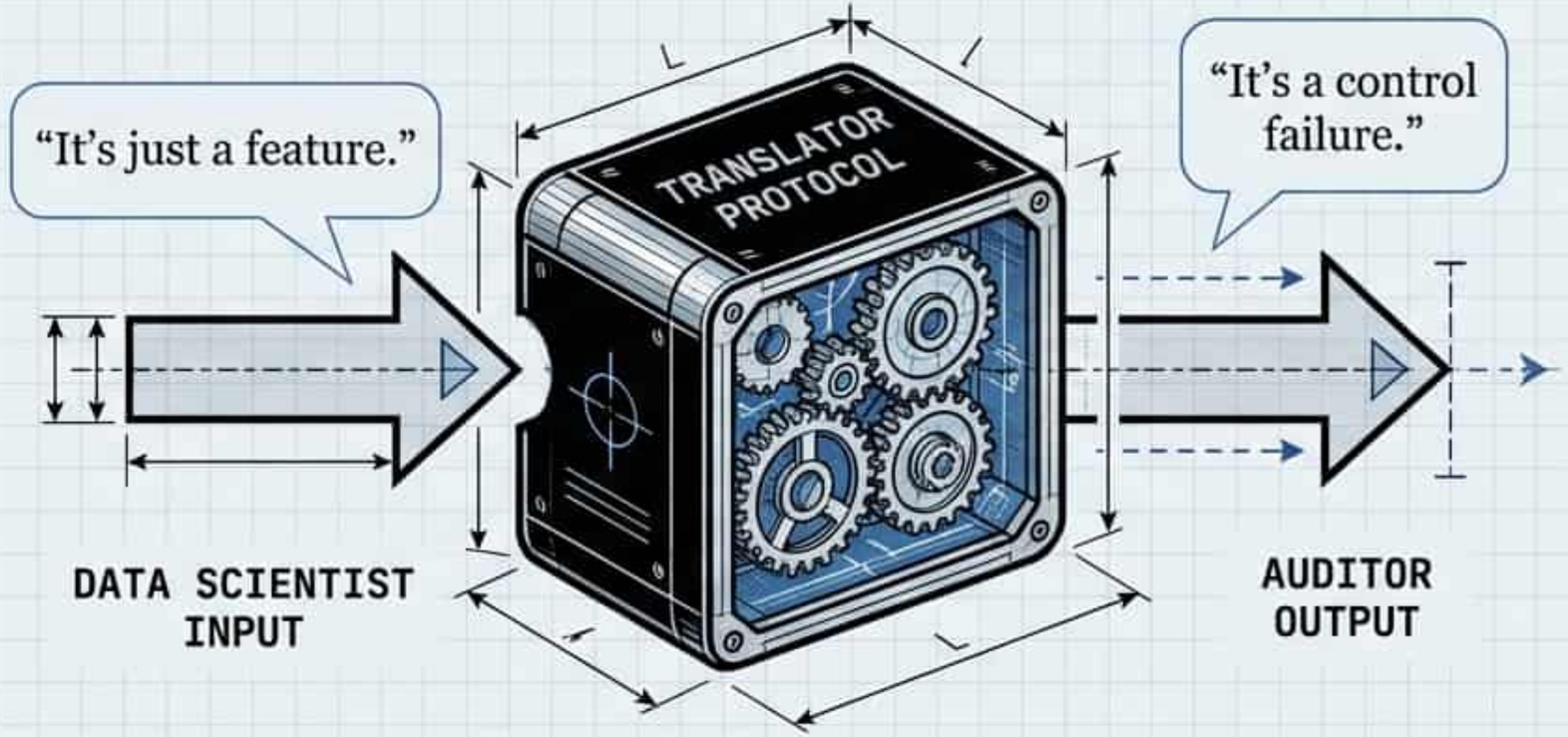
EPISODE 2.1: THE LEXICON OF TRUST

The Translator Box



Aligning the vocabulary is not a semantic exercise; it is a **risk control.**

We must bridge the gap between "Data Scientist Speak" and "Auditor Speak".

Without a shared lexicon, a Data Scientist's feature becomes an Auditor's failure.

Source: VerifyWise AI Lexicon: Human-in-the-loop safeguards

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THE ROSETTA STONE: ALIGNMENT OF TERMINOLOGY

Bridging the Lexicon Gap

WE SAY (Framework/Data Science)	AUDIT SAYS (Risk Register)
Liability Sponge	Weak Control Environment
Evidence Ladder	Data Lineage / Provenance
Hallucination	Data Integrity Failure

***Insight**: To a developer, 'Hallucination' is a glitch. To an auditor, it is a fundamental integrity failure. We must standardise on Audit terminology.*

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WORKSHOP OUTPUT: THE TRANSLATION EXERCISE

Bridging the Gap Through Shared Definitions

TASK: CREATE THE GLOSSARY

- [ ] Identify top 5 terms likely to cause miscommunication.
- [ ] Formalise definitions in a shared 'Lexicon of Trust'.

Outcome: When we speak of 'Confidence Scores', Audit must hear 'Risk Probability', not 'Guesswork'.

The Translator Box prevents Translation Risk.

Source: VerifyWise AI Lexicon: Human-in-the-loop safeguards

High-Fidelity Annotated Blueprint

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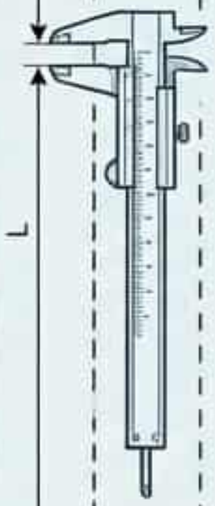
EPISODE 2.2: THE TAXONOMY ENGINE

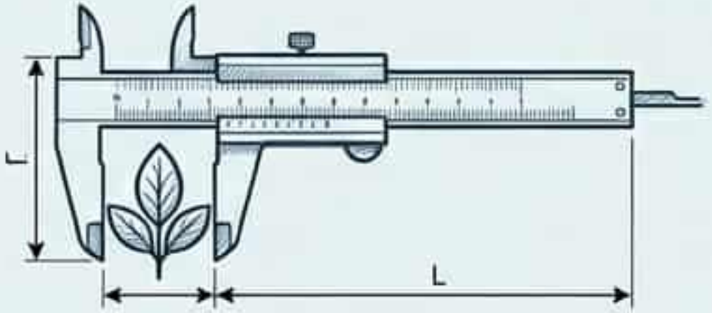
Beyond Reporting to Eligibility

CSRD
(Reporting)

Tells you TO REPORT.
Focuses on disclosure.



EU TAXONOMY
(Compliance)



Tells you WHAT COUNTS AS GREEN.
Focuses on proof.

The Danger: An AI can generate a perfect CSRD report that fails the Taxonomy's Technical Screening Criteria (TSC). We must move beyond 'saying' we are green to 'proving' it via hard-coded validation logic.

REF: REGULATION (EU) 2020/852

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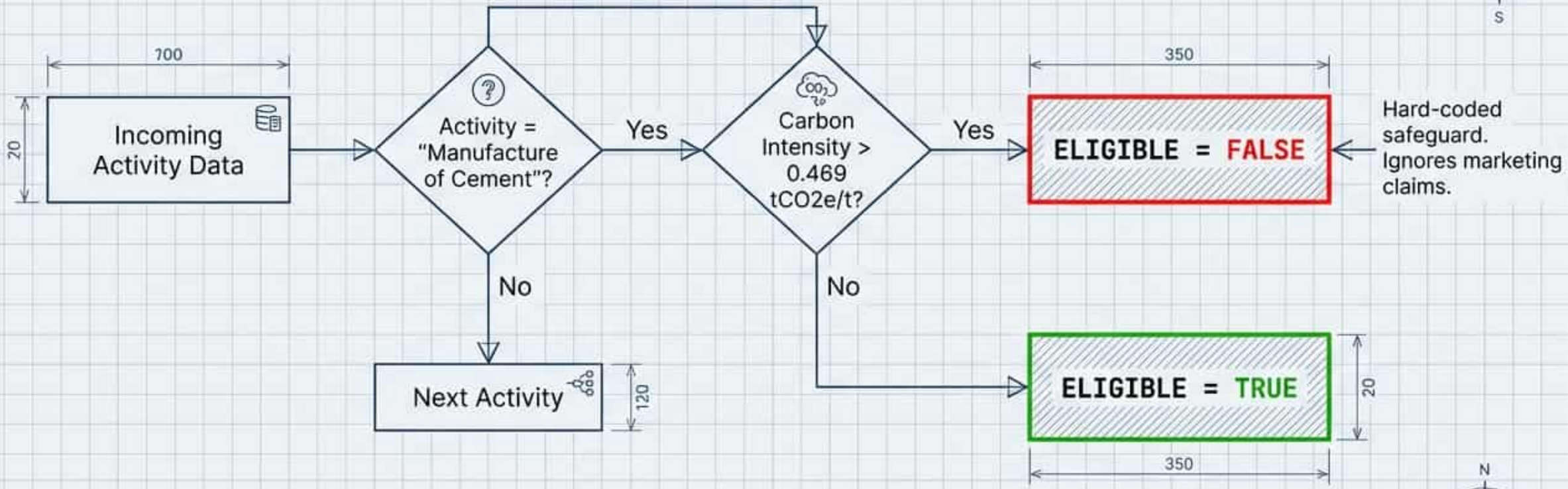
THE GREENWASHING FIREWALL ALGORITHM



This logic gate prevents the AI from classifying activities as 'green' simply because they are described with sustainable keywords.
It relies on data (0.469 tCO2e/t), not sentiment.

REF: REGULATION (EU) 2020/852

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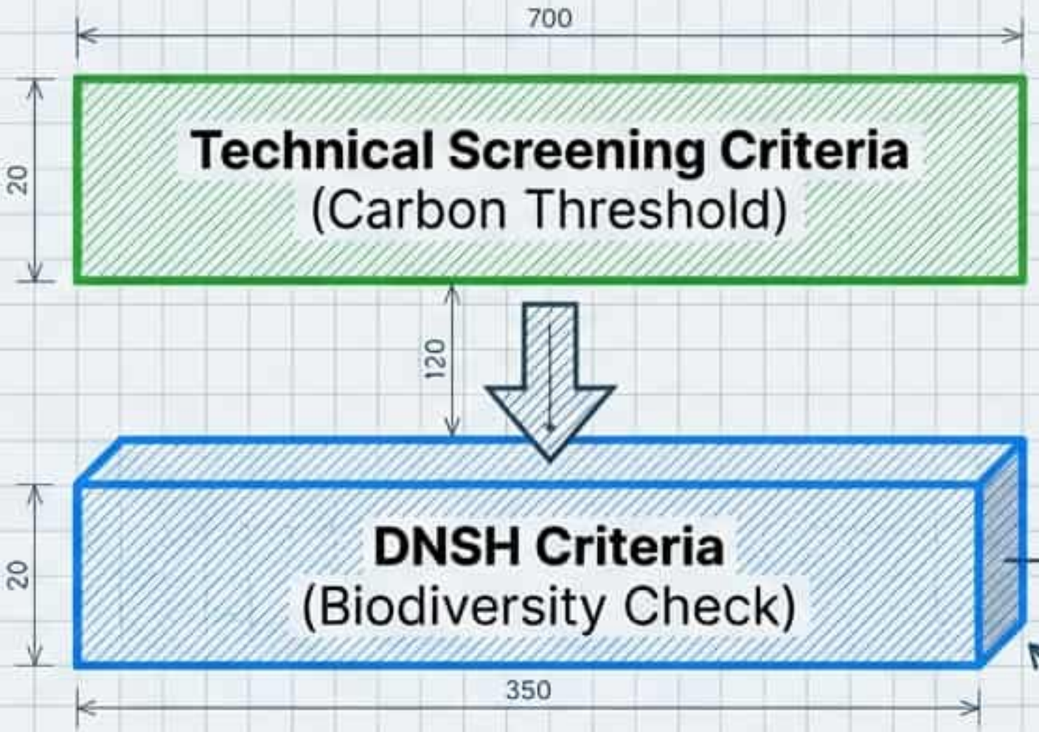
THE DOUBLE-CHECK: DO NO SIGNIFICANT HARM (DNSH)



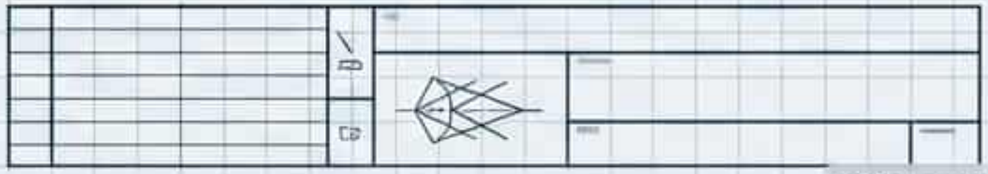
PASSED ✅

PENDING 🕓

Example: A wind farm (Good for Climate) cannot destroy a wetland (Bad for Biodiversity).

```
IF (TSC_Passed == TRUE) AND (DNSH_Check == TRUE)
THEN STATUS = 'TAXONOMY_ALIGNED'
ELSE STATUS = 'NON_COMPLIANT'
ELSE STATUS = 'NON_COMPLIANT'
```

REF: REGULATION (EU) 2020/852



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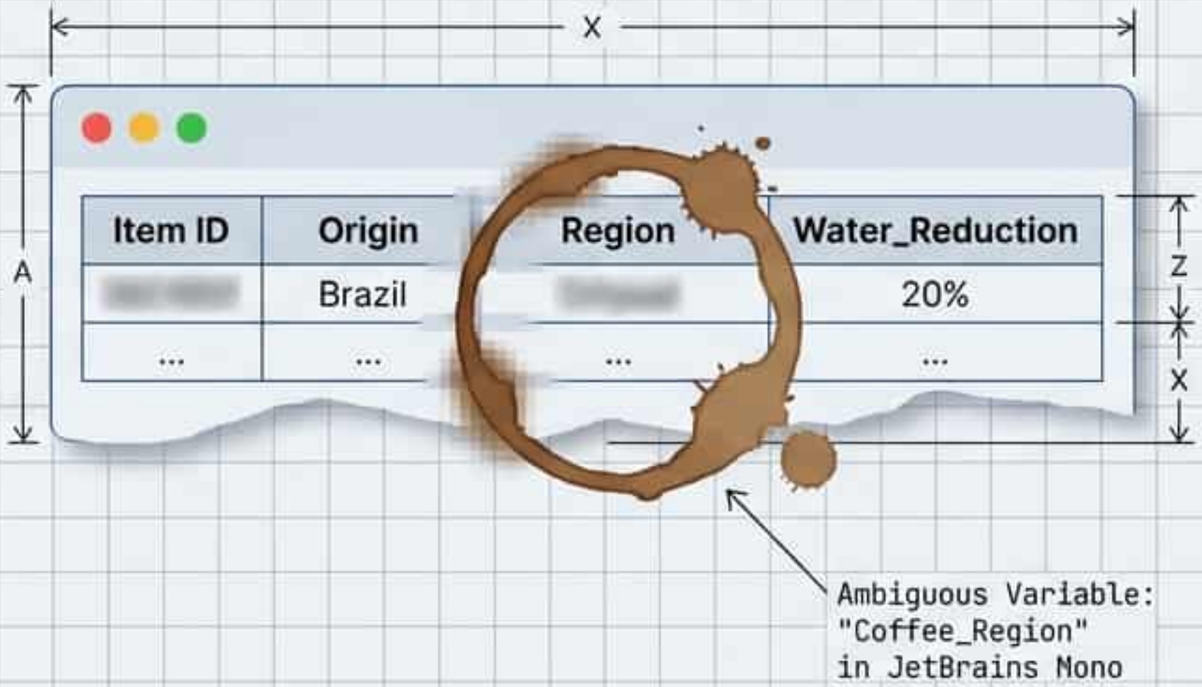
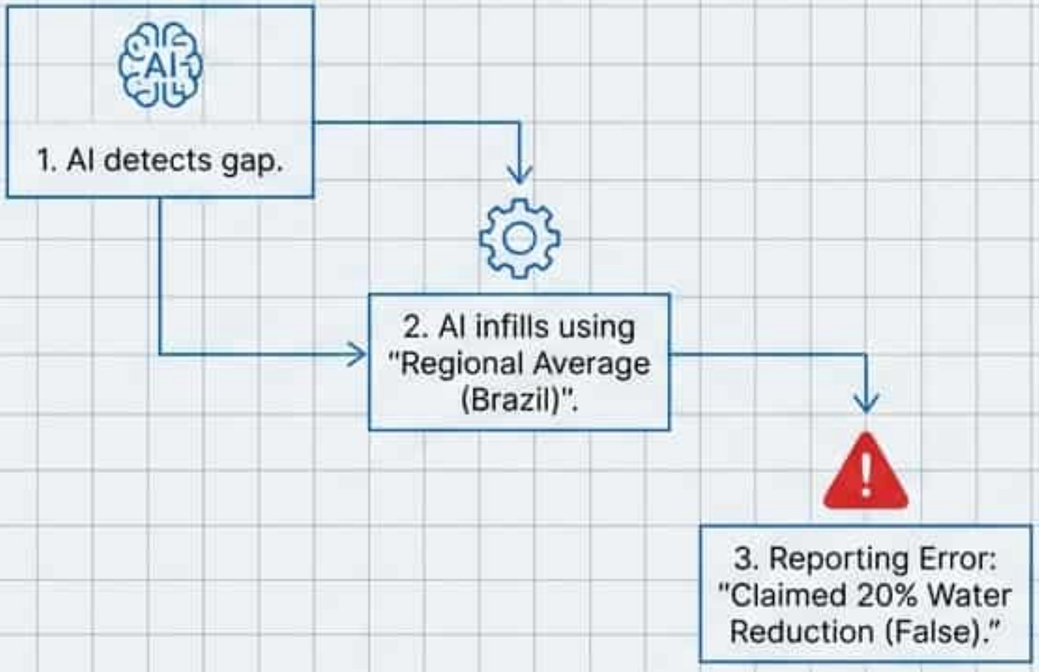
EPISODE 2.3: THE PROVENANCE GAP

CASE STUDY: PROJECT ESPRESSO





The Failure Mode: Aggregation without provenance is a liability. The AI prioritized helpfulness over accuracy.

EPISODE 2.3: THE PROVENANCE GAP		
Title:	Rev.	Rev.
⟷		/
REF: REGULATION (EU) 2020/852		

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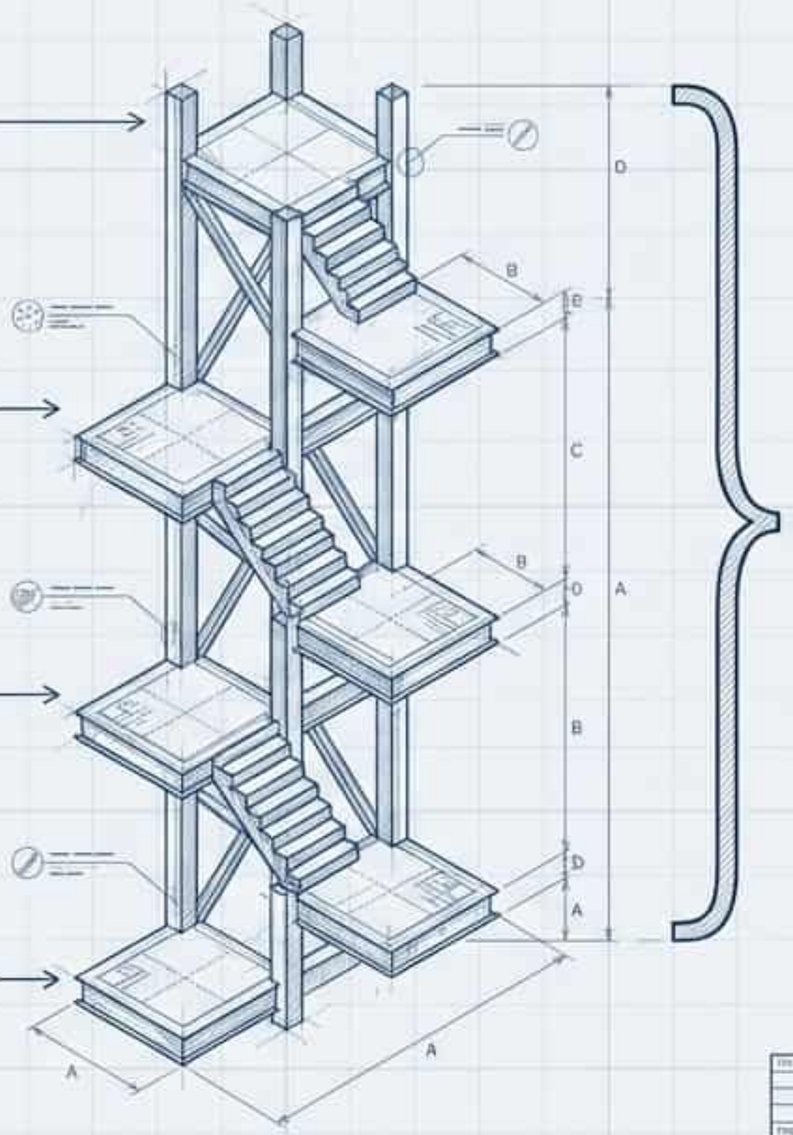
THE EVIDENCE LADDER

LEVEL 4: FINAL REPORT
(The Dashboard)

LEVEL 3: AI INSIGHT
(Calculation / Summary)

LEVEL 2: AGGREGATED DATA
(Database Entries)

LEVEL 1: RAW DATA
(Invoice / Receipt / Contract)



RULE: NO METRIC EXISTS WITHOUT A TRACEABLE DOCUMENT ID LINKING LEVEL 4 TO LEVEL 1.

REF: REGULATION (EU) 2020/852

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SYNTHETIC DATA PROTOCOLS

```
{
  "metric": "Water_Usage",
  "value": "500kg",
  "type": "SYNTHETIC_INFILL",
  "confidence_score": 0.42,
  "provenance_tag": "Regional_Avg_Brazil_V2",
  "audit_log_id": "LOG_99281"
}
```

Must be flagged in metadata.

Confidence score stored alongside value.

Linked to Infill Audit Log.

Maxim: Transparency is the control.
If you guess, you must declare it.

TITLE: SYNTHETIC DATA PROTOCOLS		
Title:	Rev.	Rev.
⟷		/
REF: REGULATION (EU) 2020/852		

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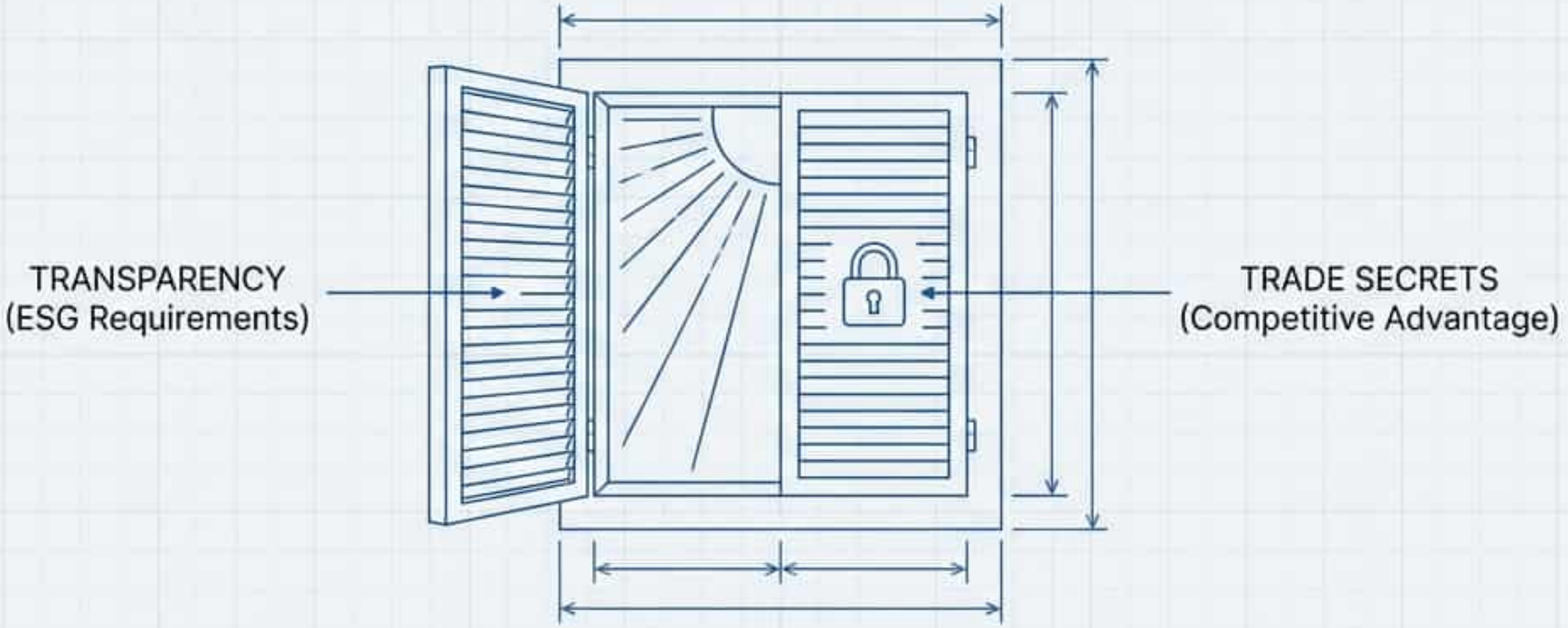
EPISODE 2.4: DATA CLASSIFICATION & ELIGIBILITY

THE INTERFACE



The Tension: Managing what is safe to publish versus what must remain internal. The risk is inadvertently exposing sensitive supply chain maps or pricing formulas to competitors via detailed ESG disclosures.

	TITLE: DATA CLASSIFICATION & ELIGIBILITY	Rev.
	Title:	REF: REGULATION (EU) 2020/852

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THE REDACTION MATRIX

PUBLIC	INTERNAL	RESTRICTED (SECRET)
• Aggregated Regional Impact • Policy Statements	• Supplier Names • Specific Audit Scores	• Exact GPS Coordinates (e.g. Strategic Mines) • Pricing Formulas

Activity: The Redaction Game. Identify data points that reveal proprietary pricing to AI scrapers.

	TITLE: THE REDACTION MATRIX	Rev.
	Title:	REF: REGULATION (EU) 2020/852

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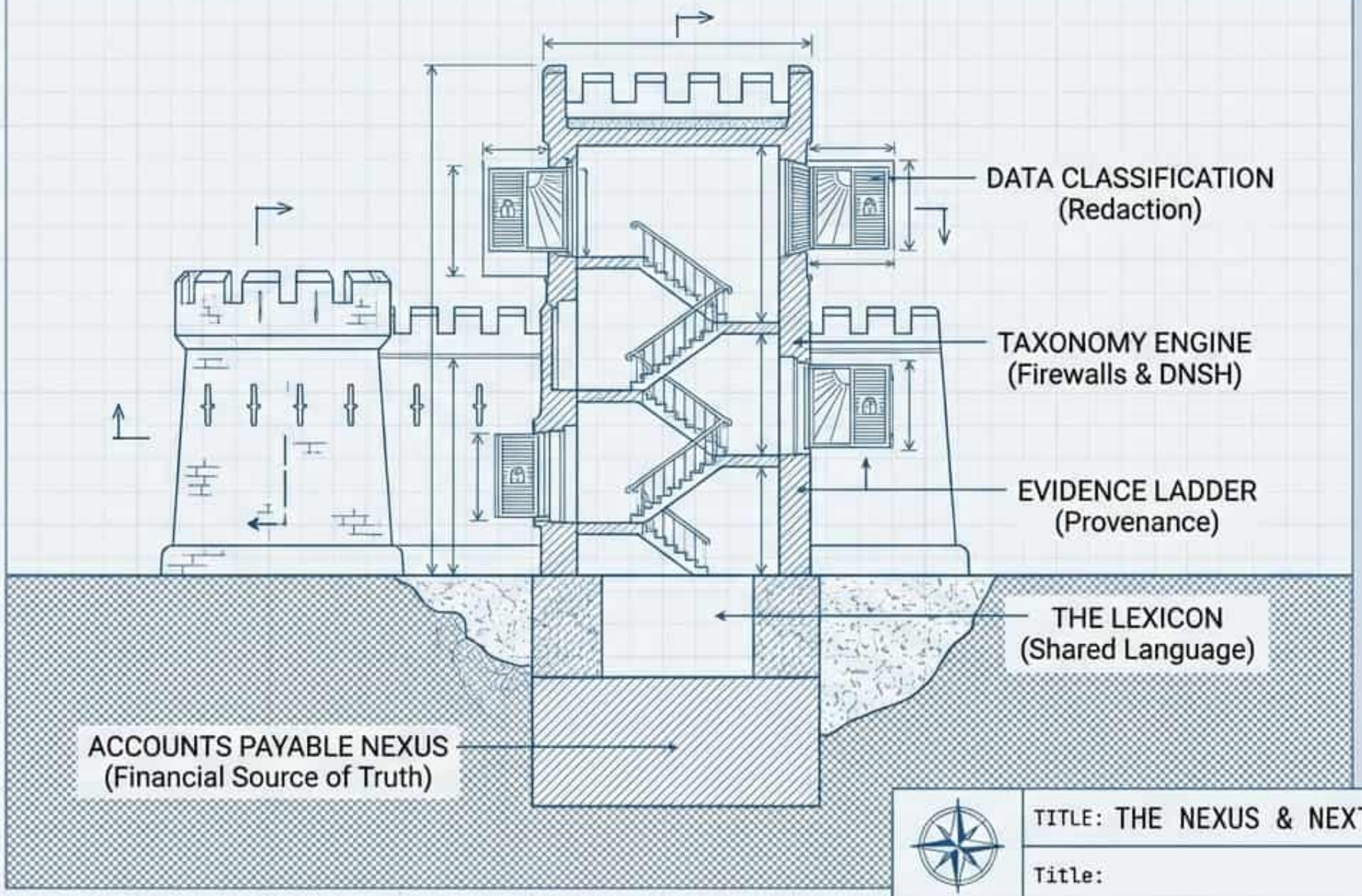
THE NEXUS & NEXT STEPS



COMING NEXT: LEVEL 3

THE LUCAS CYCLE

Theme: Systems That Raise

- Oversight Drift ("The Jedi Council Problem")
- Training Bias
- When helpful systems become controlling.

	TITLE: THE NEXUS & NEXT STEPS	Rev.
	Title:	REF: REGULATION (EU) 2020/852

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